

Session Objective

Understand what software is, the main software categories, the functions of system software and operating systems, and the basics of programming languages. Learn essential technical vocabulary.

1. Software

Definition

Software is a collection of programs and instructions that tell a computer how to perform tasks. It allows the hardware to operate.

In simple terms: Software is everything you cannot touch.

Types of Software

Type	Description	Examples
System Software	Controls hardware and runs core functions	OS, Drivers, BIOS
Application Software	Helps users perform tasks	VLC, WhatsApp, PowerPoint
Utility Software	Maintains and protects the system	Backup tools, Disk cleanup

Definitions of Each Type

System Software manages hardware and provides an environment for all programs. Examples: Operating systems, drivers, firmware.

Application Software helps users perform tasks such as browsing, writing, or editing. Examples: Firefox, AutoCAD, VLC.

Utility Software performs maintenance and security tasks. Examples: Compression tools, backup software, disk cleanup tools.

2. System Software

System software includes all programs that control basic computer functions:

- Managing hardware
- Running essential tasks
- Allowing other software to function

The most important system software is the **Operating System**.

3. Operating System (OS)

An Operating System manages computer resources such as hardware, software, files, memory, processes, and user interaction.

In simple terms: The OS is the “boss” of the computer.

Key Roles of an Operating System

1. Process Management – controls programs and CPU time
2. Memory Management – allocates and monitors RAM
3. File System Management – organizes files and permissions
4. Device Management – communicates with hardware
5. Security and User Management – controls access
6. User Interface – GUI or CLI interaction

Types of Operating Systems

Type	Description	Examples
Single-user OS	One user at a time	Windows 11 Home, macOS Ventura
Multi-user OS	Multiple users simultaneously	Linux Ubuntu Server, UNIX
Mobile OS	For smartphones and tablets	Android 14, iOS 17
Network OS	Manages network resources	Windows Server 2022, RHEL

4. Programming Languages

A programming language allows programmers to write instructions that computers can execute.

In simple terms: It is the language used to command the computer.

1. Low-level Languages

Examples: Assembly, Machine Code

Low-level languages work very close to the hardware. They offer maximum control but are difficult to read.

Uses:

- Hardware control
- Embedded systems
- Microcontrollers
- Firmware
- Performance optimization

Simple explanation: Low-level languages give full control over hardware but are hard to understand.

2. High-level Languages

Examples: Python, Java, C#, C++

High-level languages are easier to read and write. They hide hardware complexity.

Uses:

- Software development
- Desktop applications
- Mobile applications
- Games
- Artificial Intelligence

Simple explanation: High-level languages are easier to use and suitable for complex applications.

3. Web Languages

Examples: HTML, CSS, JavaScript

- HTML: Structure of the webpage
- CSS: Design and layout
- JavaScript: Interactivity

Uses: Websites, web applications, user interfaces.

4. Database Languages

Example: SQL

SQL is used to manage and query databases.

Uses:

- Websites and applications
- Banking systems
- Social media
- Cloud computing

Simple explanation: Database languages store, manage, and retrieve large amounts of data.

5. Useful Vocabulary

Term	Definition	Example
Run	Start a program	Run the application
Install	Add a program	Install the driver
Update	Improve software	Update your OS
Debug	Fix errors	Debug the code
Compile	Convert to machine code	The code failed to compile
Execute	Perform instructions	The OS executes tasks
Algorithm	Steps to solve a problem	Sorting algorithms

Acronyms

- SQL – Structured Query Language
- HTML – HyperText Markup Language
- CSS – Cascading Style Sheets
- JS – JavaScript